Week 5 Part E — written justification

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Channel budget. In D1, x-position is log GDP per capita and y-position is bounded 0–100 healthcare access because these are the two values readers compare most precisely. Bubble area carries population with the declared 18–360 points² mapping and real-value legend; hue plus six marker shapes identify region. The $120,219 GDP outlier is retained, labelled and shown on the required log axis. D2 uses five percentile-rank axes for relative association, grey background profiles and one orange regional brush; D3 uses sequential light-to-dark blue for ordered incidence. vizlib.worst_pair measures the regional palette and the figure uses redundant shapes/labels rather than claiming hue alone solves CVD.

Parallel coordinates. geolib.corr_chain() sets the axis order before plotting: healthcare_access → cases_per_100k → urban_pct → median_age → gdp_pc. The adjacent access/incidence relationship is r=-0.799; cases/urban is r=-0.619. Rank scaling shows relative order, not original gaps. This makes the multivariate path/crossing visible while retaining raw endpoint labels.

Spatial decisions. D3 maps cases per 100,000 residents (not counts), uses five natural/Jenks classes with printed breaks, and projects all districts with one equal-area reference latitude so polygon area is not exaggerated by latitude. The 24 code joins all succeed; the boundary is the synthetic fictional federation geometry distributed for CDB601220, with extraction date 2026-11-09. No unit was suppressed. Counts return in a sorted companion bar chart.

Counterfactual and ethics. A raw-count choropleth would darken D17, D13 and D09, whose residents generate more cases in absolute terms; the normalized-rate leaders are D19, D20 and D21, with zero top-three overlap. Counts correlate with population at r=+0.986, so the raw map would mainly point leadership toward populous districts, not the highest per-resident incidence. Week 5's required five-reader activity has not been run; its blank results file is a capture template, not evidence, so no claimed reader mean is used here. Small-area estimates are separately checked against Poisson limits, and no low-count unit is presented as certain.